ASSIGNMENT REPORT – AUTOMATIC CONTROL SYSTEMS

INDOOR HYDROPONIC GREENHOUSE CONTROL USING PID

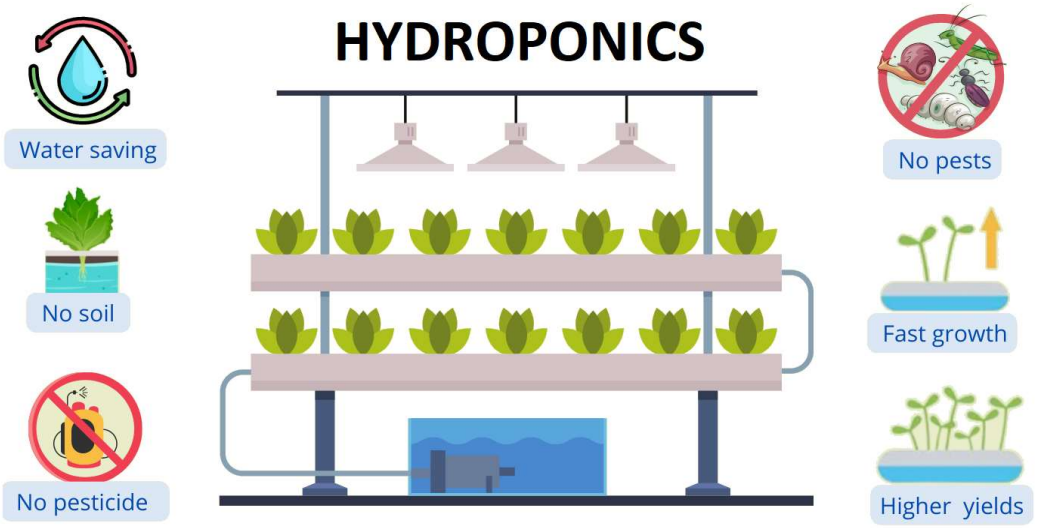


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COURSE TITLE : AUTOMATIC CONTROL SYSTEMS
COURSE CODE : 19R402
DATE : 24/03/2023

HYDROPONICS

Hydroponics is a type of horticulture and a subset of hydroculture which involves growing plants, usually crops or medicinal plants, **without soil, by using water-based mineral nutrient solutions** in aqueous solvents. Terrestrial or aquatic plants may grow with their roots exposed to the nutritious liquid or in addition, the roots may be mechanically supported by an inert medium such as perlite, gravel, or other substrates.



Hydroponic greenhouse control typically involves monitoring and controlling several environmental variables, such as pH level, temperature, humidity, and nutrient levels, to ensure optimal plant growth. Here are some of the sensors generally used in hydroponic greenhouse control:

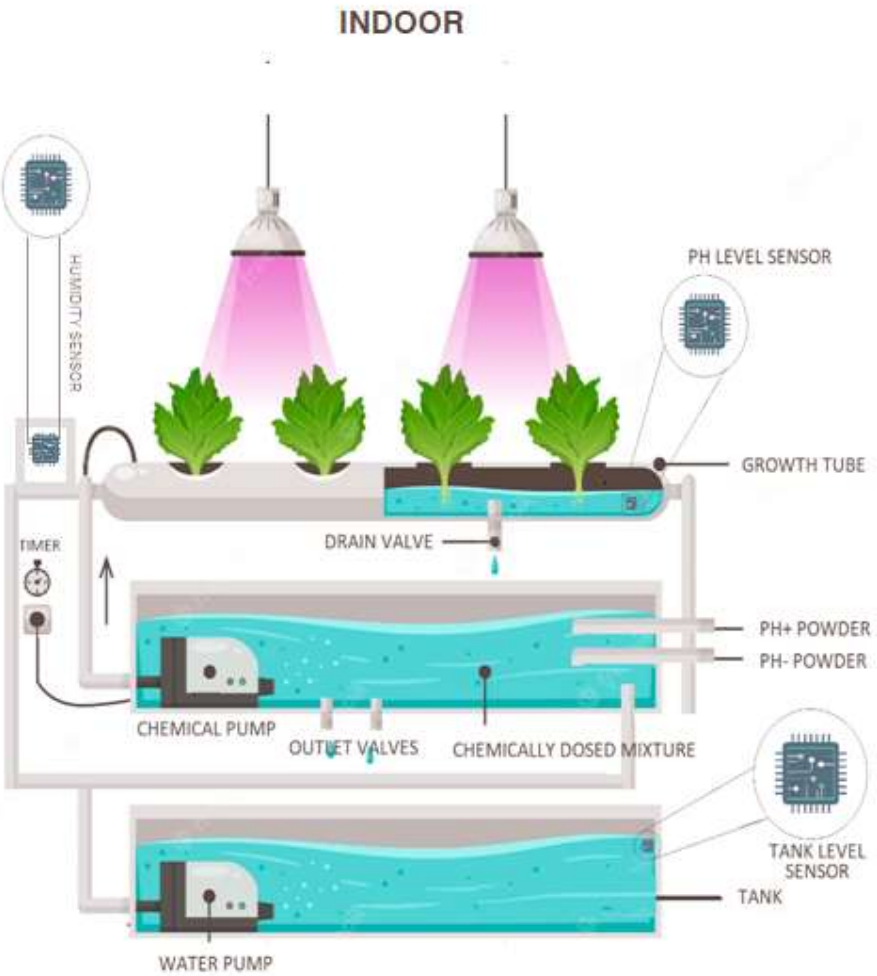
- **pH sensor**: Measures the pH level of the nutrient solution in the hydroponic system.
- **Temperature sensor**: Measures the temperature of the air or nutrient solution in the hydroponic system.
- **Humidity sensor**: Measures the humidity level of the air in the hydroponic system.
- **Electrical conductivity sensor**: Measures the nutrient concentration in the hydroponic solution.

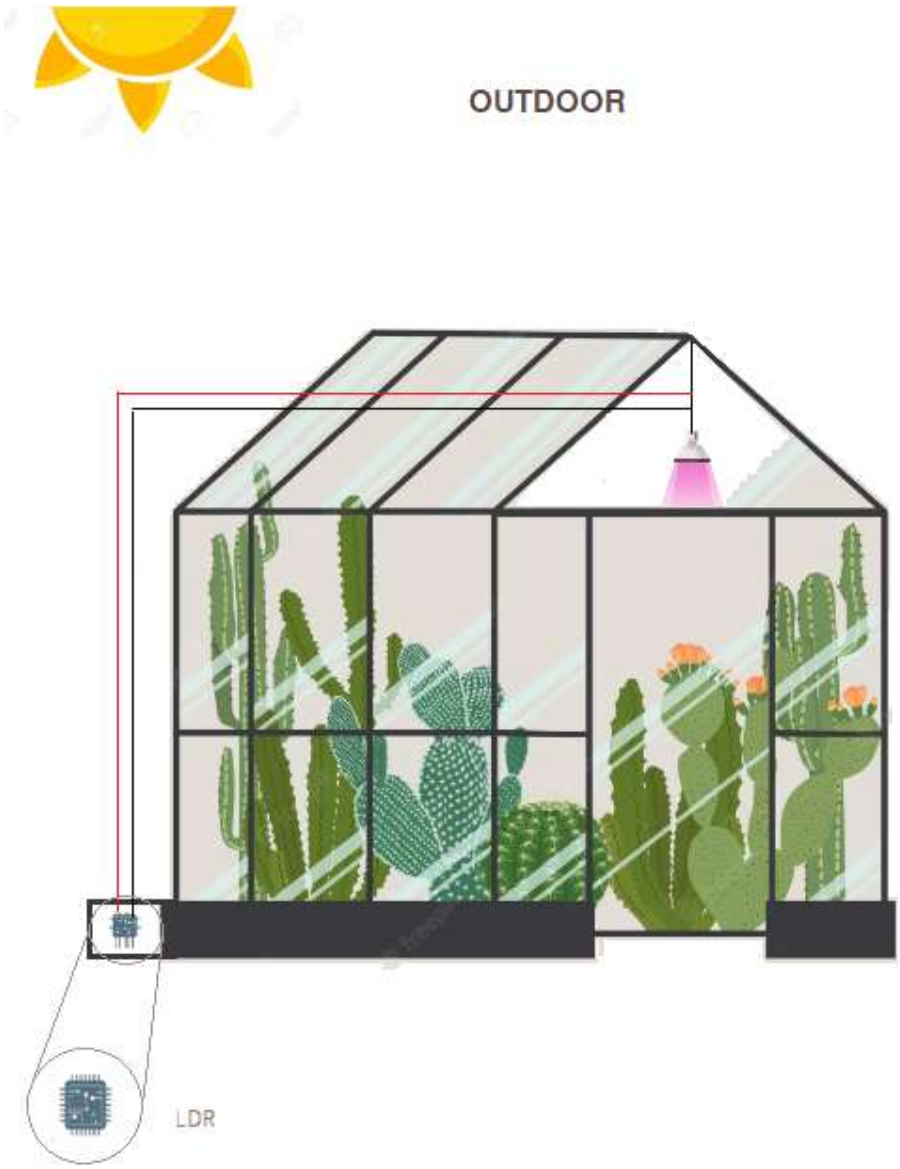
- **Dissolved oxygen sensor**: Measures the amount of oxygen dissolved in the hydroponic solution.

- **Light sensor**: Measures the intensity of light in the hydroponic system.

These sensors are typically connected to a microcontroller or a computer-based control system that uses algorithms to maintain optimal environmental conditions for plant growth. PID controllers are often used to control the pH level and other variables in hydroponic greenhouse systems.

SCHEMATIC DIAGRAM



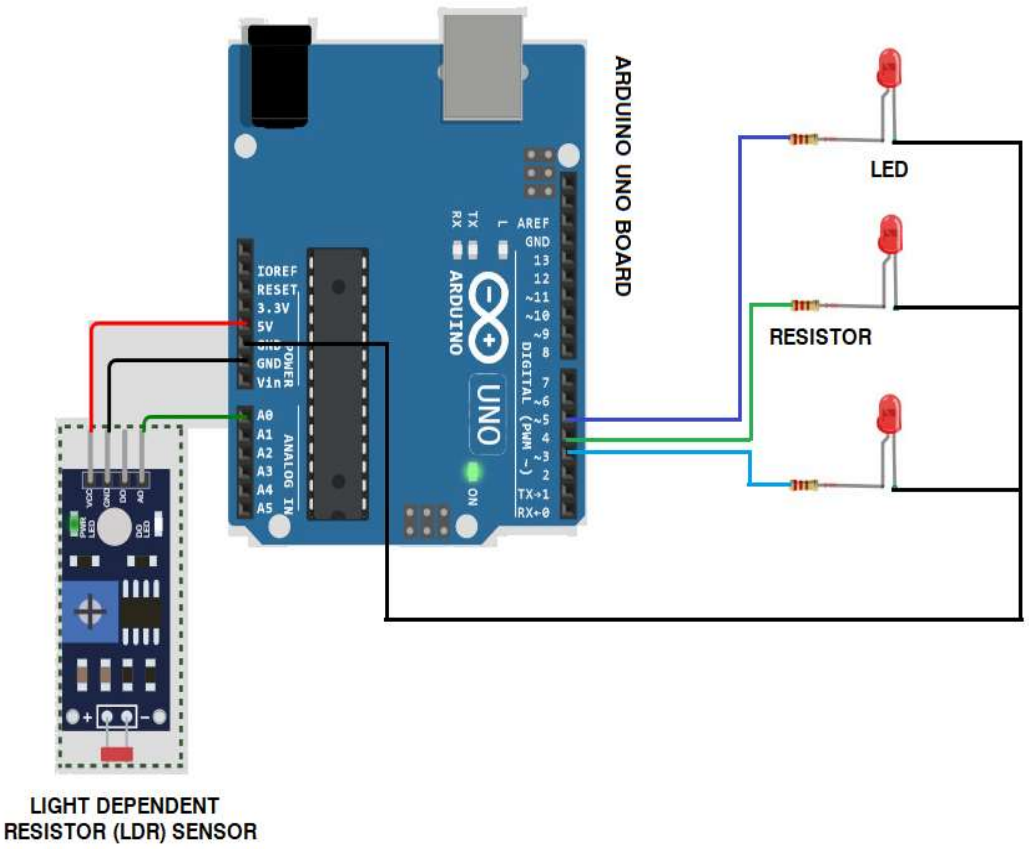


HARDWARE : PID CONTROL OF LIGHT INTENSITY INDOORS

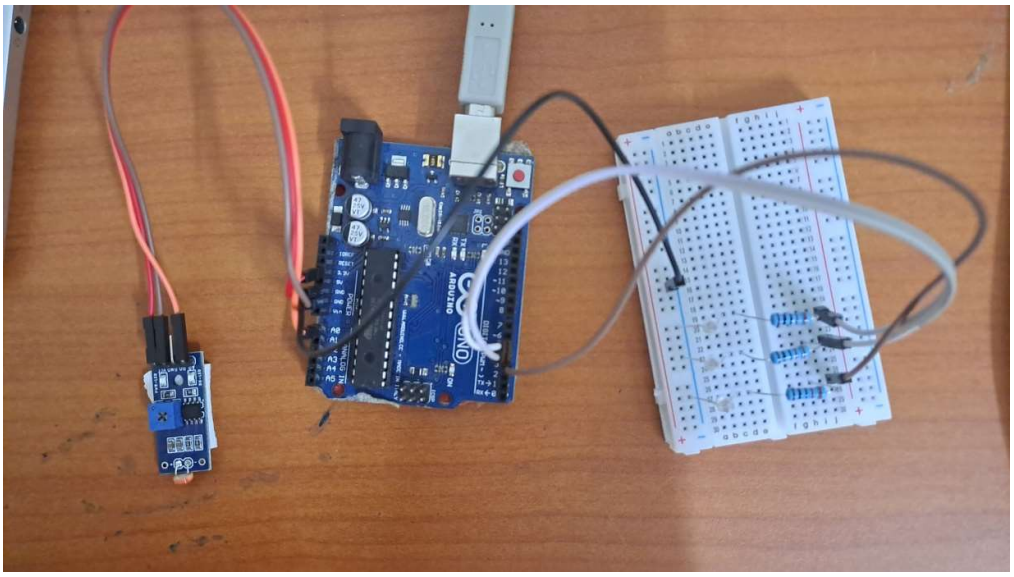
COMPONENTS REQUIRED

1.	Arduino Uno Board	x	1
2.	LDR sensor module	x	1
3.	LED	x	3
4.	Resistor (220 ohms)	x	3
5.	Breadboard	x	1
6.	Jumper wires	x	as required

CIRCUIT



PHYSICAL CONNECTIONS



CODE

```
#include <PID_v1.h>
#define PIN_INPUT 0
#define PIN1_OUTPUT 3
#define PIN2_OUTPUT 5
#define PIN3_OUTPUT 6
//Define Variables we'll be connecting to
double Setpoint, Input, Output;
//Define the aggressive and conservative Tuning Parameters
double aggKp=4, aggKi=0.2, aggKd=1;
double consKp=1, consKi=0.05, consKd=0.25;
//Specify the links and initial tuning parameters

PID myPID(&Input, &Output, &Setpoint, consKp, consKi, consKd, DIRECT);

void setup()
{ //initialize the variables we're linked to
  Input = analogRead(PIN_INPUT);
  Setpoint = 100;
  Serial.begin(9600);
  //turn the PID on
  myPID.SetMode(AUTOMATIC);}

void loop()
{  Input = map(analogRead(PIN_INPUT), 0, 1023, 0, 255);
  double gap = abs(Setpoint-Input); //distance away from setpoint

  if (gap < 10)
  {  //we're close to setpoint, use conservative tuning parameters
    myPID.SetTunings(consKp, consKi, consKd);  }

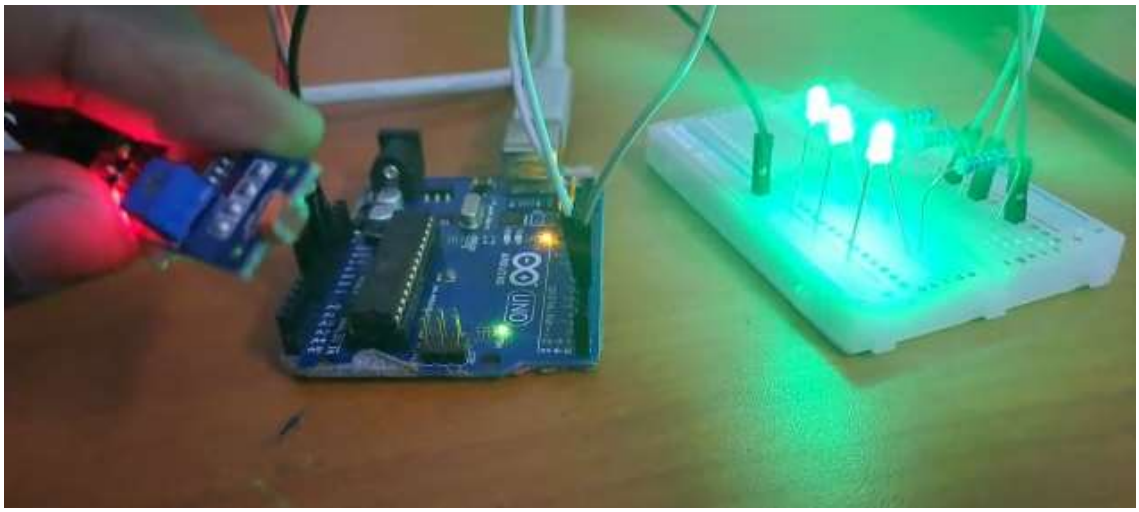
 else
  { //we're far from setpoint, use aggressive tuning parameters
     myPID.SetTunings(aggKp, aggKi, aggKd); }
  myPID.Compute();
```

```
analogWrite(PIN1_OUTPUT, Output);
analogWrite(PIN2_OUTPUT, Output);
analogWrite(PIN3_OUTPUT, Output);
Serial.print("LDR reading: ");
Serial.print(Input);
Serial.print(", Output: ");
Serial.println(Output); }
```

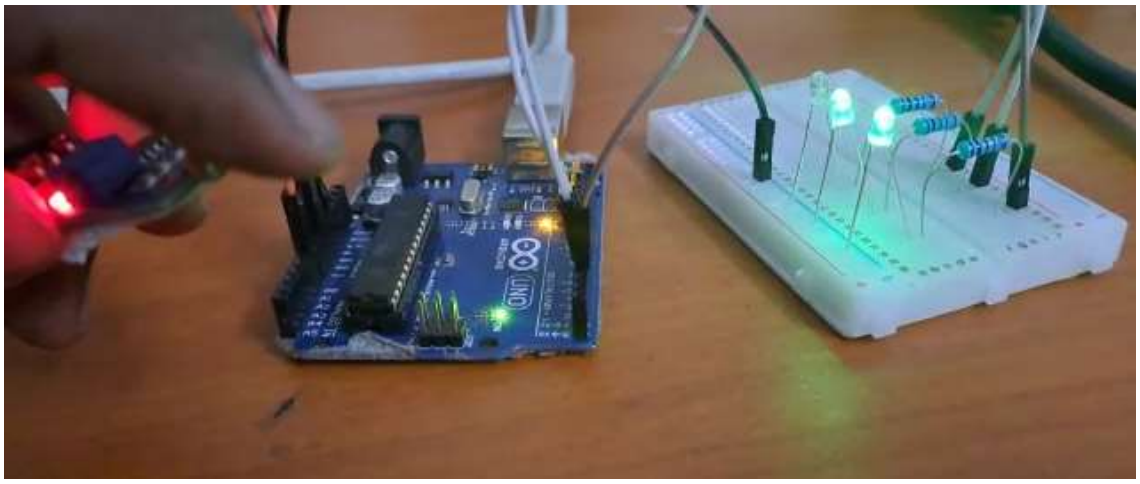
WORKING

With the usage of PID (Proportional Integral Derviative) controller concept, we have integrated PID library in our code and manually input Kp, Ki, Kd (respective proportional, integral and derivative constants. This helps in constantly comparing the outdoor light intensity and stabilizing the indoor light intensity accordingly for the indoor hydroponic greenhouse setup. To conceptually integrate this into our application, the LDr (Light Dependent Resistor) is placed outdoor (usually in rooftops or nearby peaks) and the stabilizing happens in the light setups of indoor greenhouse rooms. This is important because different plants have different light requirements for optimal growth. For example, plants that are in the **vegetative stage** require **more blue light**, while plants in the **flowering stage** require **more red light**. So the respective lighting setups **mimic the external brightness and provide stable corresponding internal light intensity**.

OUTPUT



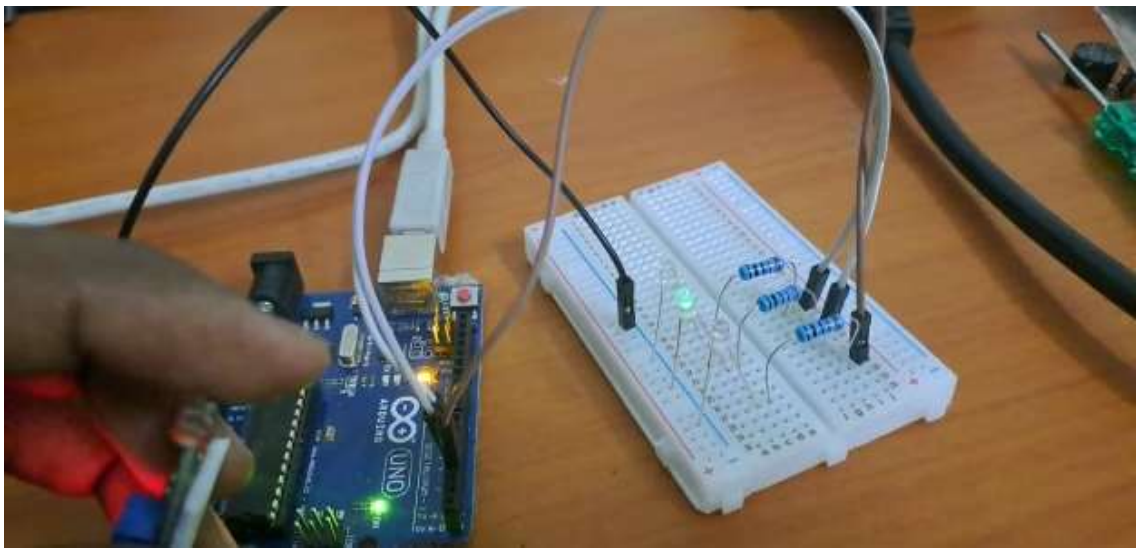
INPUT : Exposure to the environment **OUTPUT** : Full brightness capacity



INPUT : Partial exposure to the environment **OUTPUT** : Reduced brightness

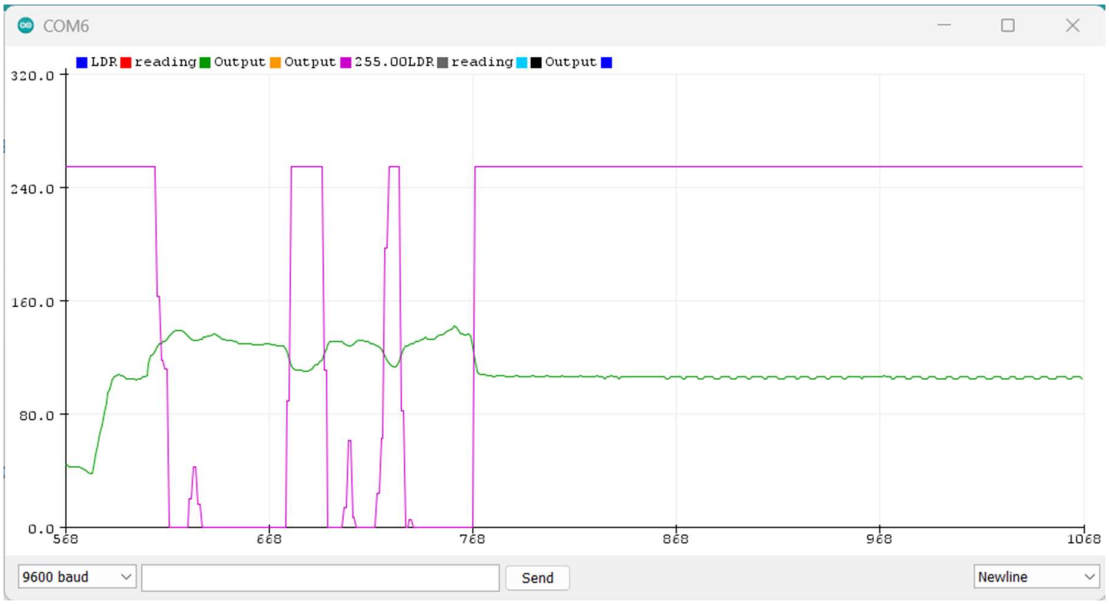


INPUT : No exposure to the environment **OUTPUT** : No brightness



INPUT : Minimal exposure to the environment **OUTPUT** : Minimal brightness

GRAPHICAL OUTPUT



BENEFITS OF LIGHTING CONTROL

Controlling the light in an indoor hydroponic greenhouse can provide several benefits, including:

- **Consistency**: By using artificial lighting, the system can ensure that the plants receive a consistent amount of light each day, regardless of the weather or time of year.
- **Efficiency**: Artificial lighting can be used more efficiently than natural light, as the system can use only the wavelengths of light that the plants need for optimal growth.
- **Flexibility**: With programmable lighting systems, the light can be adjusted to meet the specific needs of different plant species, growth stages, and environmental conditions.
- **Increased Yield**: By providing the plants with optimal light conditions, the system can increase the yield and quality of the harvest.

Overall, controlling the light in an indoor hydroponic greenhouse is an important aspect of the system that can significantly impact plant growth and yield.

SOFTWARE : PID CONTROL OF PH, TEMPERATURE AND HUMIDITY SENSOR

```
clear all;

clc;

% Set simulation parameters

setpoint = 7.0; % Setpoint pH

Kp = 1; % Proportional gain

Ki = 0.1; % Integral gain

Kd = 0.01; % Derivative gain

integralError = 0; % Initialize integral error

lastError = 0; % Initialize last error


while true

    % Generate random sensor values

    pH = rand*3 + 5; % pH range from 5 to 8

    temperature = rand*10 + 20; % temperature range from 20 to 30

    humidity = rand*20 + 40; % humidity range from 40 to 60


    % Calculate error and control signal using PID controller

    error = setpoint - pH;

    integralError = integralError + error;

    derivativeError = error - lastError;

    controlSignal = Kp*error + Ki*integralError + Kd*derivativeError;

    controlSignal = max(0, min(1, controlSignal)); % Limit the control signal to a range between 0 and 1


    % Display results in the command window

    fprintf('pH = %.2f\n', pH);

    fprintf('Temperature = %.2f\n', temperature);

    fprintf('Humidity = %.2f\n', humidity);
```

```
    fprintf('Control Signal = %.2f\n\n', controlSignal);


    % Update last error
    lastError = error;


    pause(5); % Wait for 5 seconds before generating new values
end
```

MATLAB WINDOW VALUES IN 5 SECOND INTERVAL

MATLAB Command Window Page 1

```
pH = 5.00
Temperature = 26.02
Humidity = 51.00
Control Signal = 1.00

pH = 5.79
Temperature = 29.85
Humidity = 44.45
Control Signal = 1.00

pH = 7.79
Temperature = 26.52
Humidity = 51.33
Control Signal = 0.00

pH = 7.33
Temperature = 27.47
Humidity = 56.65
Control Signal = 0.00

pH = 7.84
Temperature = 27.05
Humidity = 49.74
Control Signal = 0.00

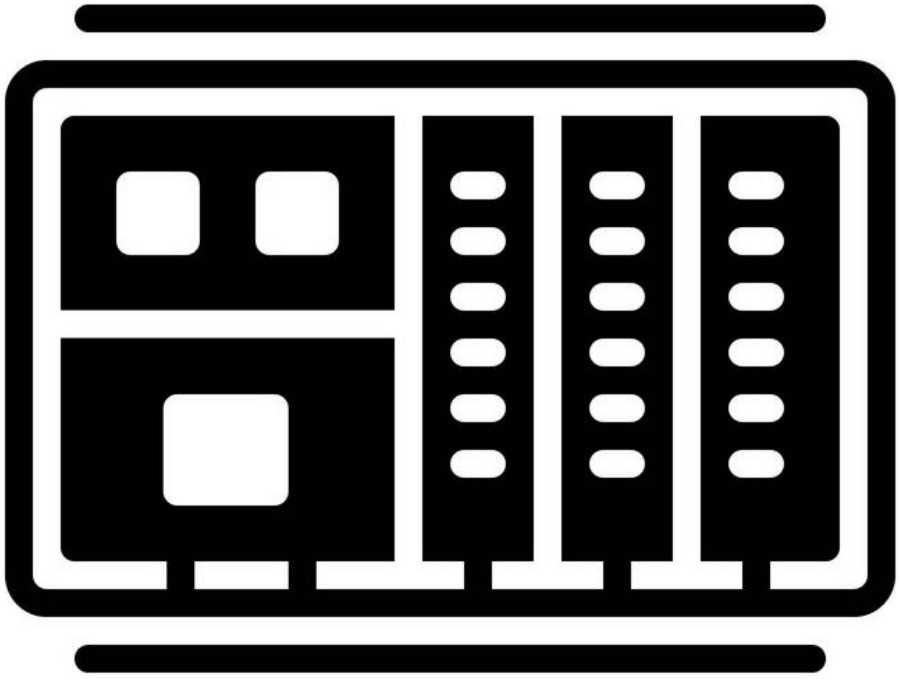
pH = 7.52
Temperature = 26.34
Humidity = 46.93
Control Signal = 0.00
```

CONCLUSION

The indoor hydroponic greenhouse system's control was implemented via hardware for light intensity control and software for control using PID mode of control. This promotes robustness, adaptability, energy efficiency and higher yields.

INDIVIUAL REPORT - PLC AND SCADA LABORATORY

AUTOMATED HYDROPONIC GREENHOUSE CONTROL USING PLC & SCADA



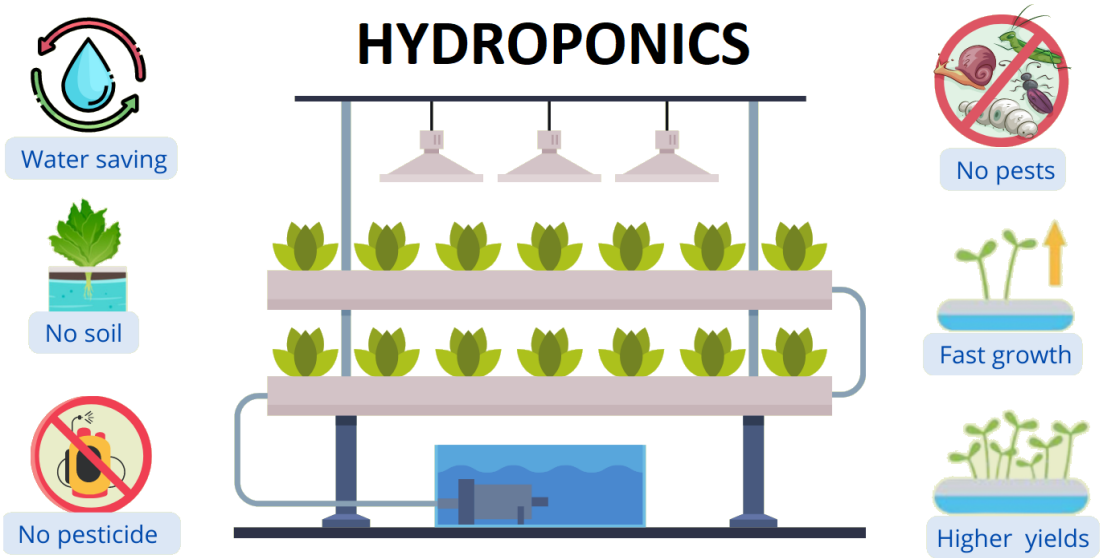
NAME : Mullainathan
ROLL NO : 22R435
COURSE TITLE : PLC AND SCADA LABORATORY
COURSE CODE : 19R412
DATE : 24/03/2023

PROBLEM STATEMENT

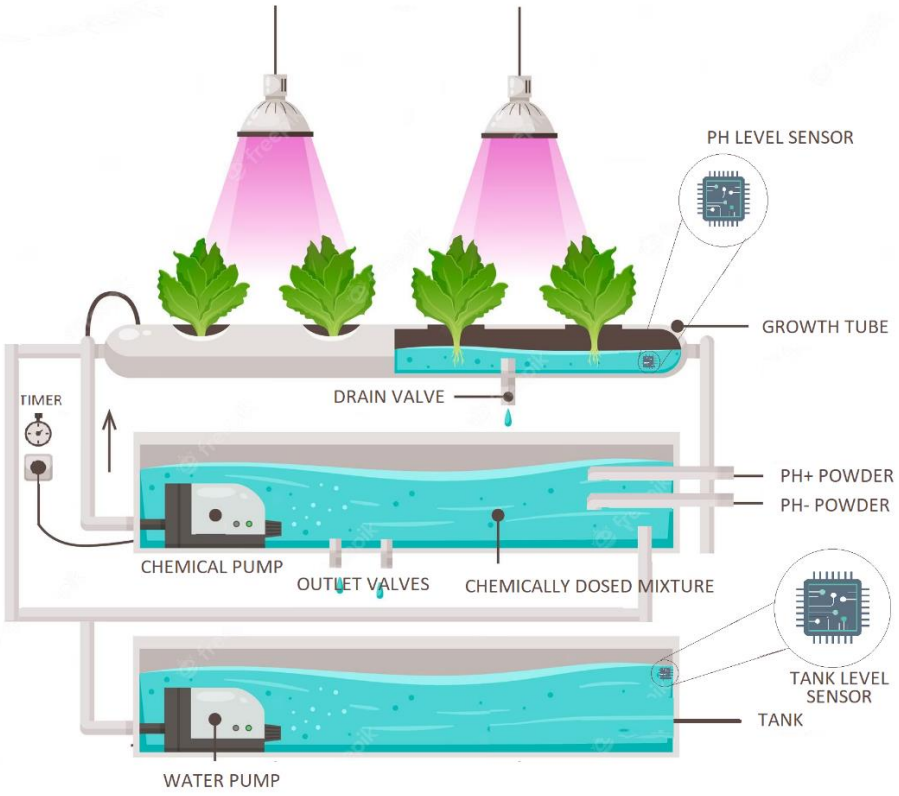
To develop a PLC ladder logic program to control and optimize the operation of an automated hydroponic greenhouse system. The system in general is to be able to monitor and control the environmental conditions to create the optimal growing conditions for various plants. In specific, the program should be able to adjust the nutrient solution and pH levels in the hydroponic system, and thus be able to automate the process of planting and watering of plants by controlling the irrigation system, nutrient dosing system, the tank's level and flow rate of the fluid mixture into the tank.

HYDROPONICS

Hydroponics is a type of horticulture and a subset of hydroculture which involves growing plants, usually crops or medicinal plants, **without soil, by using water-based mineral nutrient solutions** in aqueous solvents. Terrestrial or aquatic plants may grow with their roots exposed to the nutritious liquid or in addition, the roots may be mechanically supported by an inert medium such as perlite, gravel, or other substrates.



SCHEMATIC DIAGRAM



SOLUTION

ANALOG INPUTS

pH sensor	-	IW66, MD10
Flow rate sensor	-	IW64, MD2
Level sensor	-	Manual value input in SCADA screen, MD0

DIGITAL INPUTS

Toggle Switch	-	I0.0
Remote Switch	-	M0.0

DIGITAL OUTPUTS

Chemical Dosing Valve	-	Q0.0
PH- (PH value decrement)	-	Q0.1
PH+ (PH value increment)	-	Q0.2
Water pump	-	Q0.3
Out Valve 1	-	Q0.4
Out Valve 2	-	Q0.5

PROGRAM LOGIC

NETWORK 1

Water tank level's value is scaled from 0 – 10 feet using the scaling block that was created using FC function block with the help of NORM_X and SCALE_X blocks.

NETWORK 2

PH value is scaled from 0 – 14 using the scaling block that was created using FC function block with the help of NORM_X and SCALE_X blocks.

NETWORK 3

Under the conditions that the toggle switch (I0.0) or remote switch (M0.0) is switched ON :

- When PH value is between the range of 5 – 7, the chemical dosing valve (Q0.0) is opened with no powder mixture included.
- When PH value is greater than 7, the PH- (Q0.1) PH value decrementing powder is mixed in the chemical dosing mixture for 10 seconds and then the chemical dosing valve (Q0.0) is opened.
- When PH value is lesser than 5, the PH+ (Q0.2) PH value incrementing powder is mixed in the chemical dosing mixture for 5 seconds and then the chemical dosing valve (Q0.0) is opened.

NETWORK 4

Under the conditions that the toggle switch (I0.0) or remote switch (M0.0) is switched ON :

- When tank level lesser than 10 feet, Water pump (Q0.3) is switched ON.
- When tank level greater than 10 feet, Water pump (Q0.3) is switched OFF.

NETWORK 5

Under the conditions that the toggle switch (I0.0) or remote switch (M0.0) is switched ON :

- When flow rate is set to 0, Out Valve 1 (Q0.4) and Out Valve 2 (Q0.5) are closed.
- When flow rate is set to 50 and lesser, Out Valve 1 (Q0.4) is opened and Out Valve 2 (Q0.5) is closed.
- When flow rate is set to greater than 50, Out Valve 1 (Q0.4) and Out Valve 2 (Q0.5) are opened.

PLC LADDER LOGIC

- *TANK LEVEL SCALING FUNCTION BLOCK*

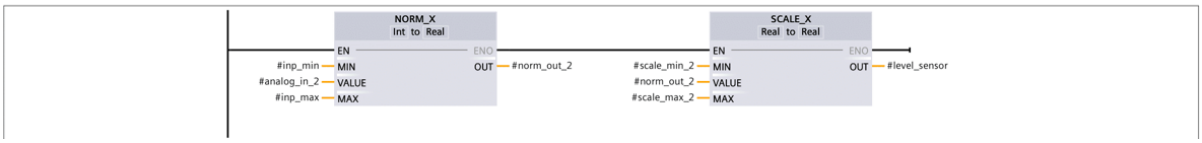
hydroponic_greenhouse / PLC_1 [CPU 1214C DC/DC/DC] / Program blocks

TANK LEVEL SCALING [FC2]

TANK LEVEL SCALING Properties							
General							
Name	TANK LEVEL SCALING	**Number**	2	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Block_1			
Name	**Data type**	**Default value**	**Comment**
Input			
analog_in_2	Int		
inp_min	Int		
inp_max	Int		
scale_min_2	Real		
scale_max_2	Real		
Output			
level_sensor	Real		
InOut			
Temp			
norm_out_2	Real		
Constant			
Return			
TANK LEVEL SCALING	Void		

Network 1:



- *PH LEVEL SCALING FUNCTION BLOCK*

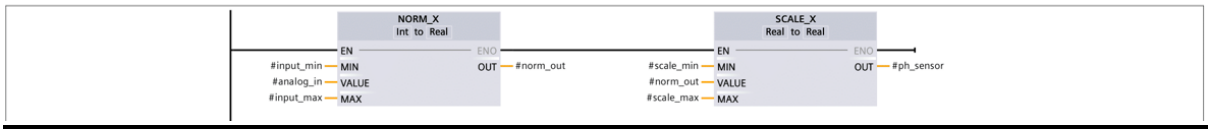
hydroponic_greenhouse / PLC_1 [CPU 1214C DC/DC/DC] / Program blocks

PH LEVEL SCALING [FC1]

PH LEVEL SCALING Properties							
General							
Name	PH LEVEL SCALING	**Number**	1	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

scaling			
Name	**Data type**	**Default value**	**Comment**
Input			
analog_in	Int		
input_min	Int		
input_max	Int		
scale_min	Real		
scale_max	Real		
Output			
ph_sensor	Real		
InOut			
Temp			
norm_out	Real		
Constant			
Return			
PH LEVEL SCALING	Void		

Network 1:



- *MAIN FUNCTION BLOCK*

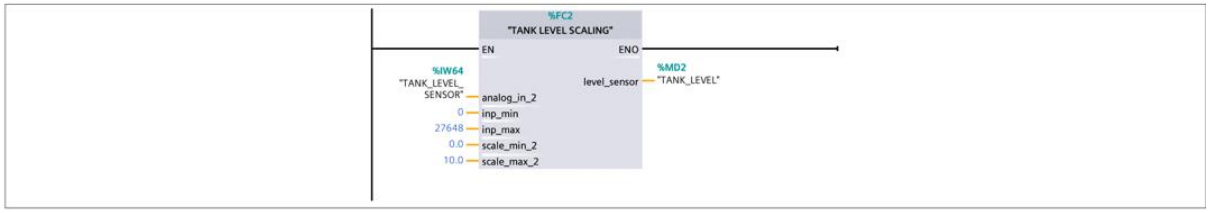
hydroponic_greenhouse / PLC_1 [CPU 1214C DC/DC/DC] / Program blocks

Main [OB1]

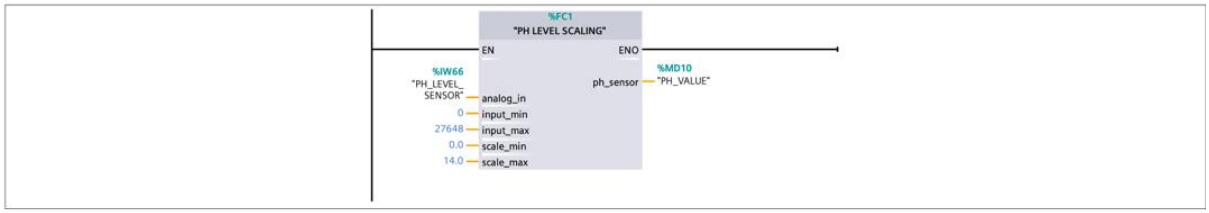
Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Main			
Name	**Data type**	**Default value**	**Comment**
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

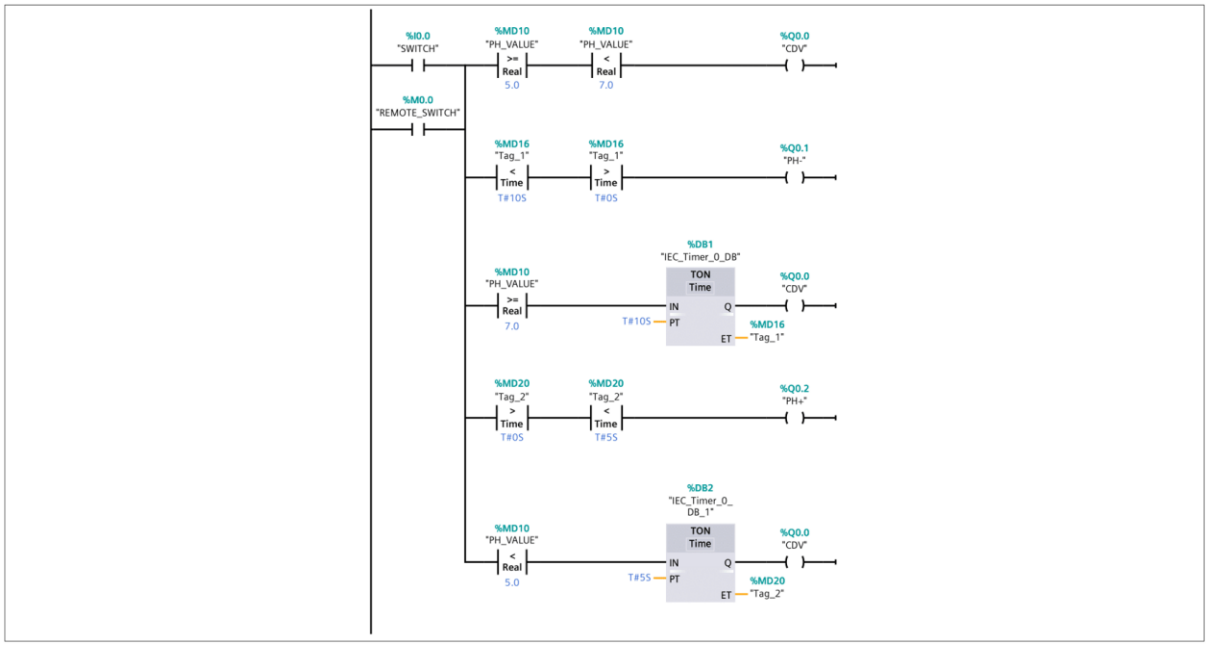
Network 1:



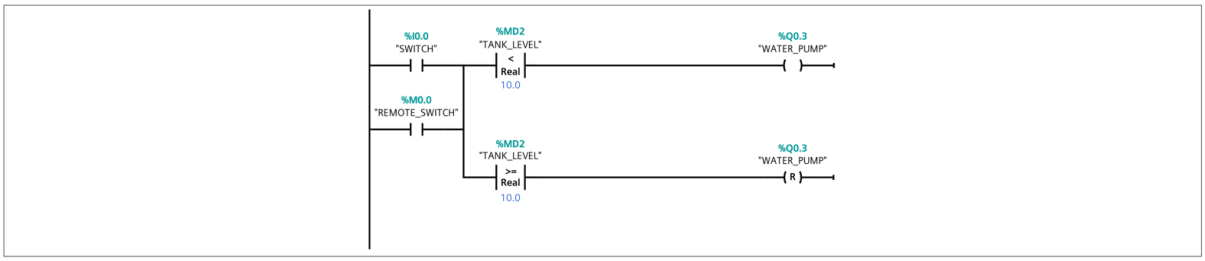
Network 2:



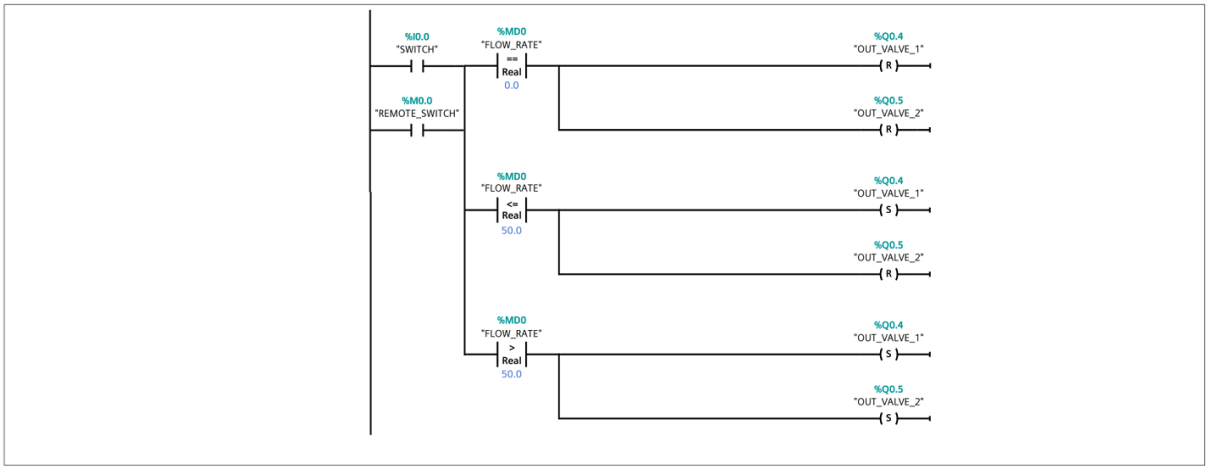
Network 3:



Network 4:

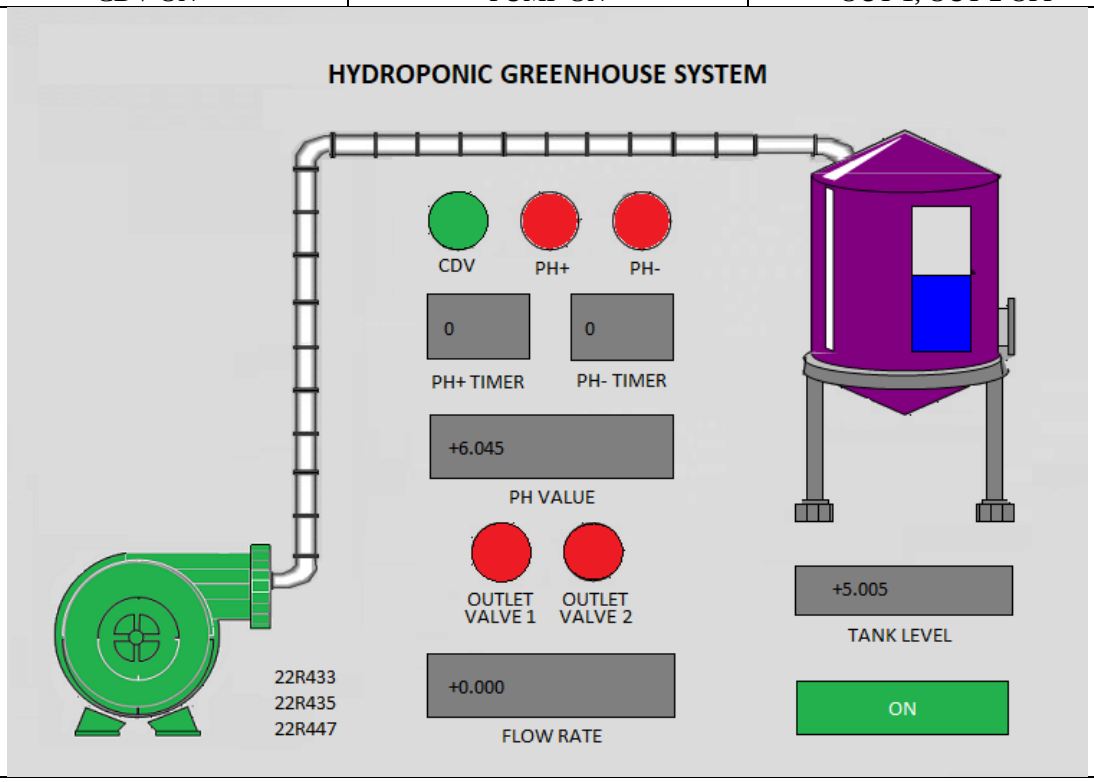


Network 5:

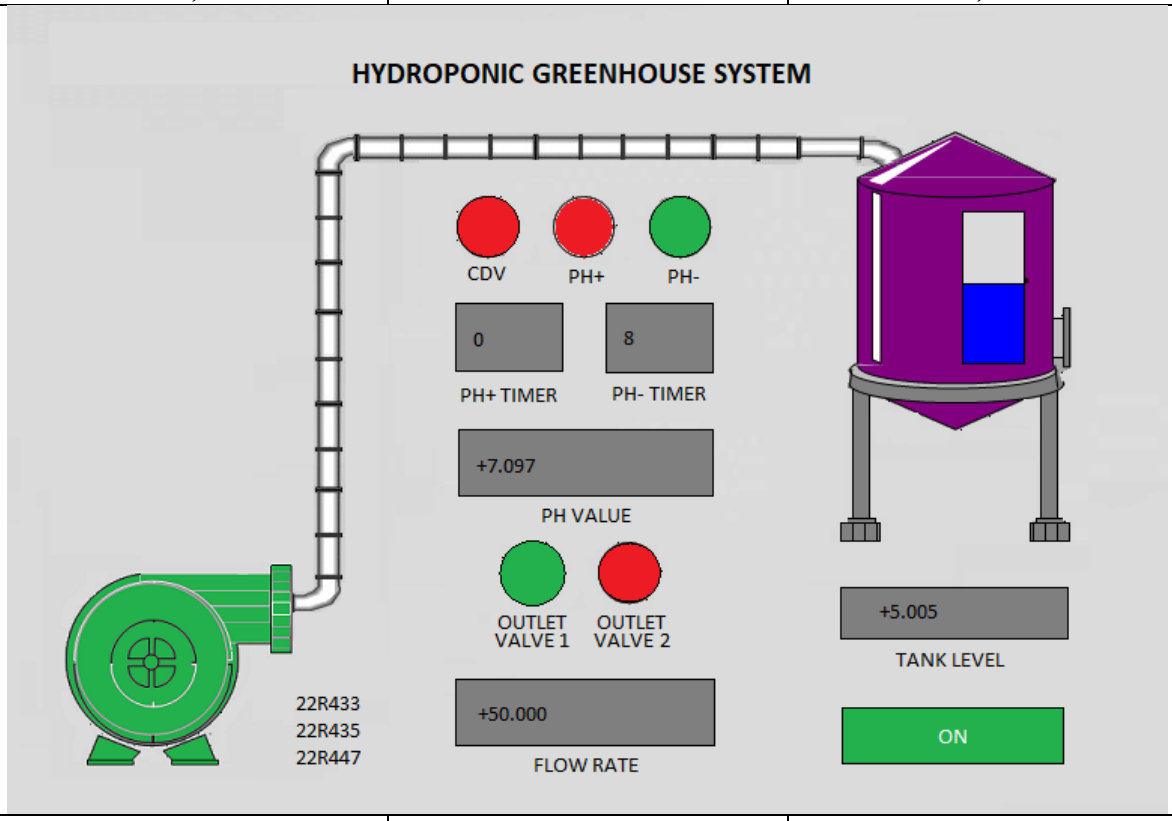


SCADA SCREEN

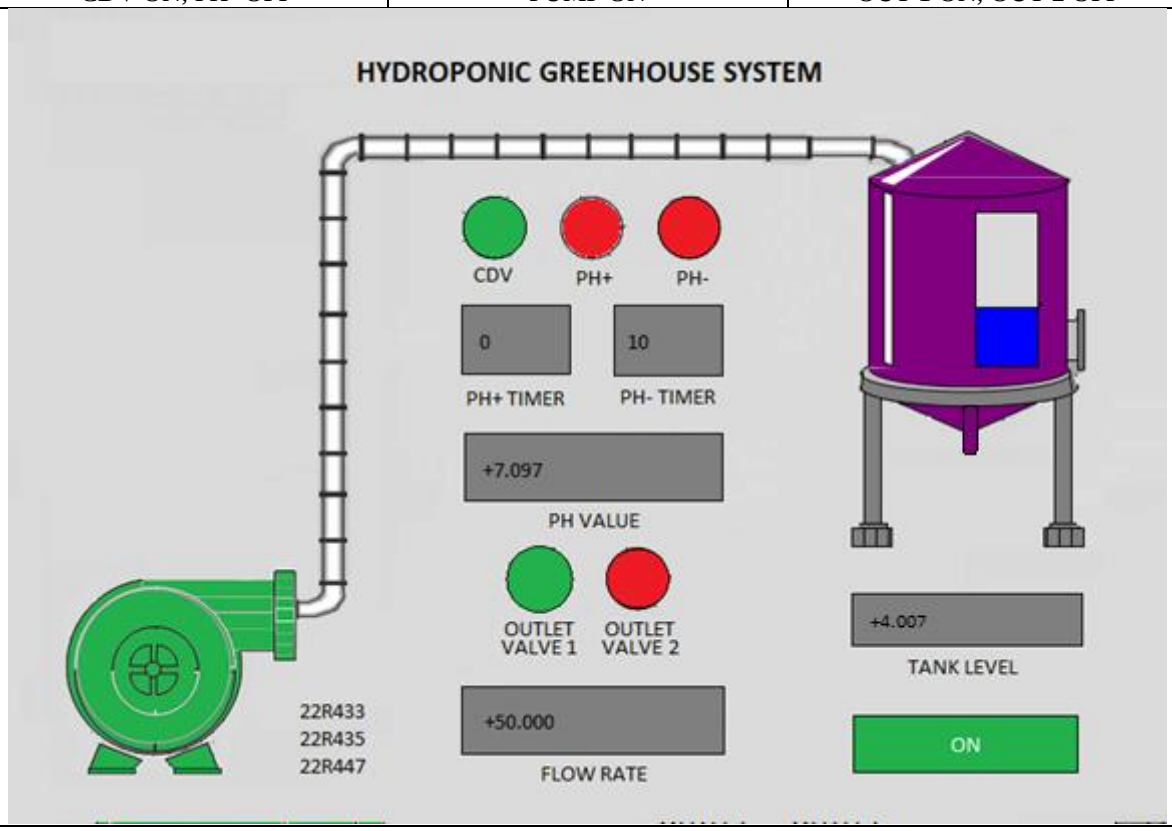
PH LEVEL - 6.045 ph	**TANK LEVEL** – 5.005 ft	**FLOW RATE** - 0
CDV ON	PUMP ON	OUT 1, OUT 2 OFF



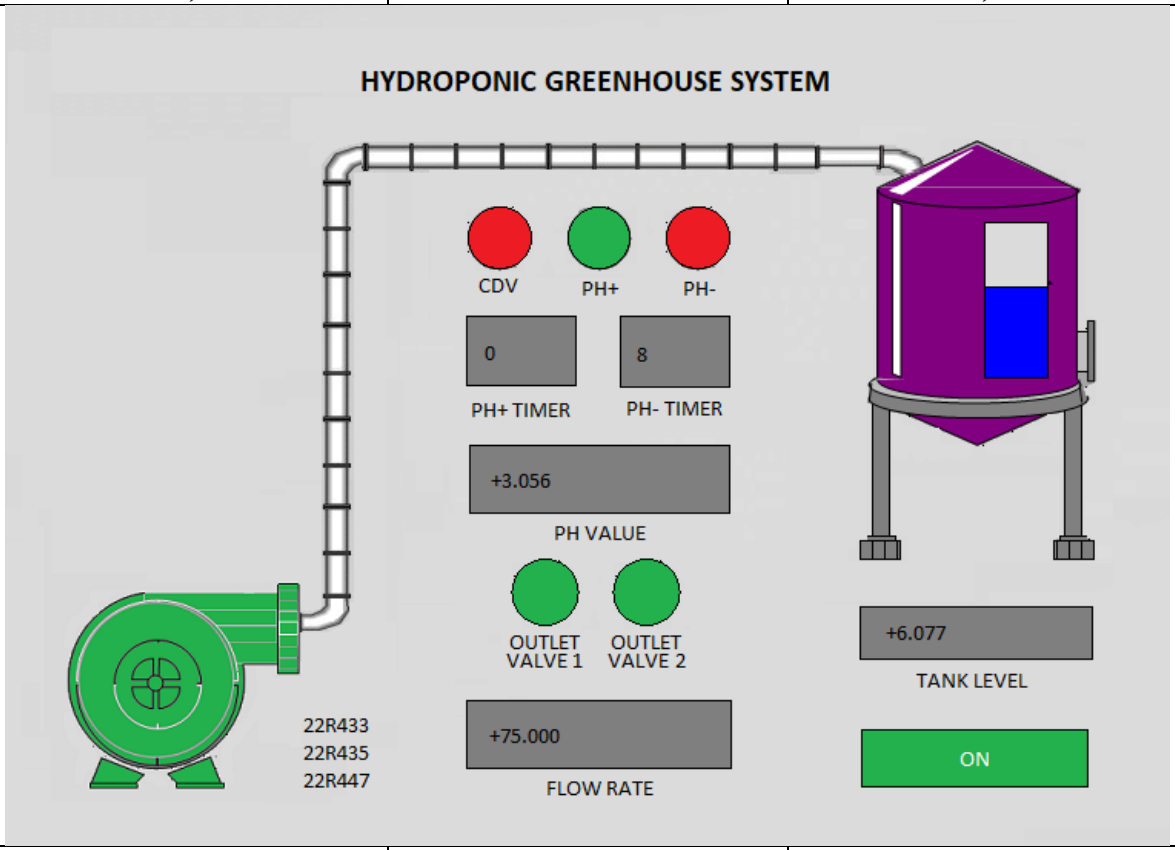
PH LEVEL – 7.097 ph	**TANK LEVEL** – 5.005 ft	**FLOW RATE** - 50
CDV OFF, PH- ON	PUMP ON	OUT 1 ON, OUT 2 OFF



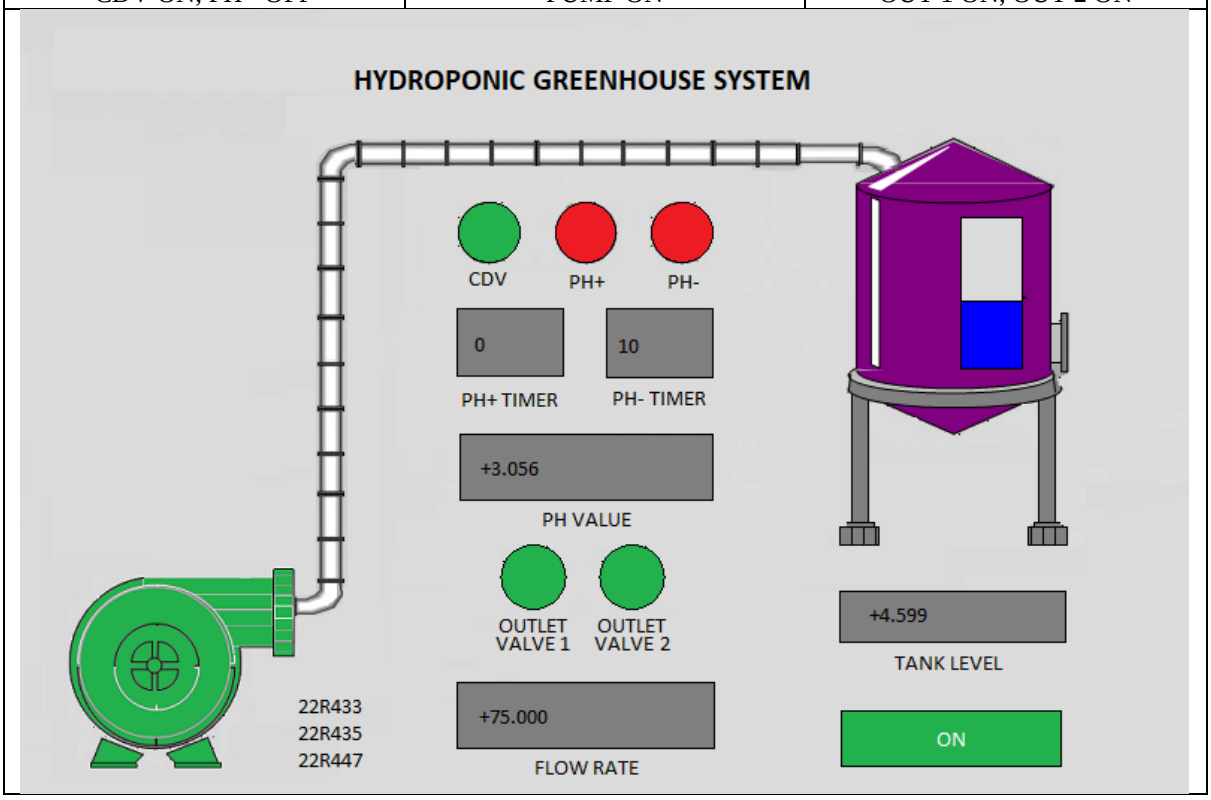
PH LEVEL – 7.097 ph	**TANK LEVEL** – 5.005 ft	**FLOW RATE** - 50
CDV ON, PH- OFF	PUMP ON	OUT 1 ON, OUT 2 OFF



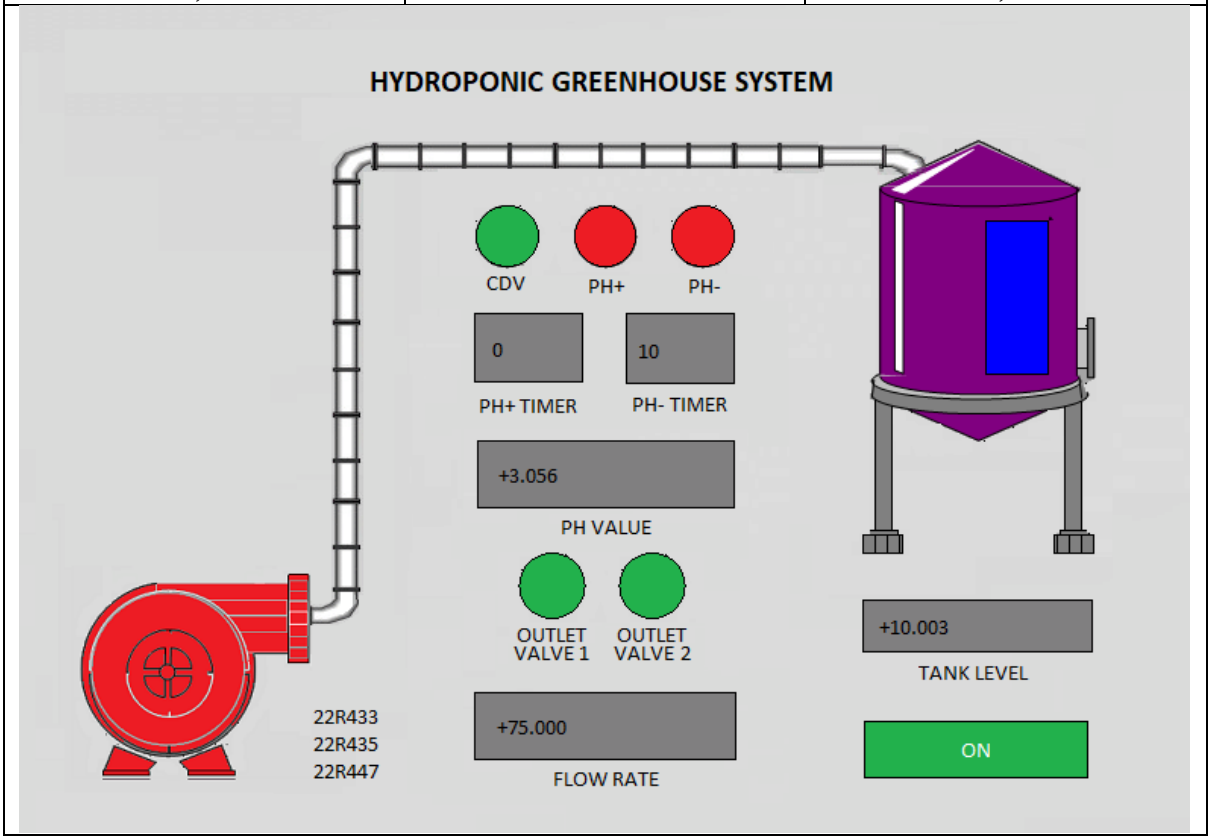
PH LEVEL – 3.056 ph	**TANK LEVEL** – 6.077 ft	**FLOW RATE** - 75
CDV OFF, PH+ ON	PUMP ON	OUT 1 ON, OUT 2 ON



PH LEVEL – 3.056 ph	**TANK LEVEL** – 6.077 ft	**FLOW RATE** - 75
CDV ON, PH+ OFF	PUMP ON	OUT 1 ON, OUT 2 ON



PH LEVEL – 3.056 ph	**TANK LEVEL** – 10.003 ft	**FLOW RATE** - 75
CDV ON, PH+ OFF	PUMP OFF	OUT 1 ON, OUT 2 ON



CONCLUSION

The automated hydroponic greenhouse system's pH level, tank level and flow rate were adjusted, and so, the respective chemical dosing, pumping, PH level increment, decrement and outlet valve's opening and closing were controlled using PLC and SCADA successfully. This results in optimization of water, chemicals, nutrients usage and enhanced plant growth in the greenhouse.